

# Exploring Trustful AI Augmentation with Virtual Evacuation Study

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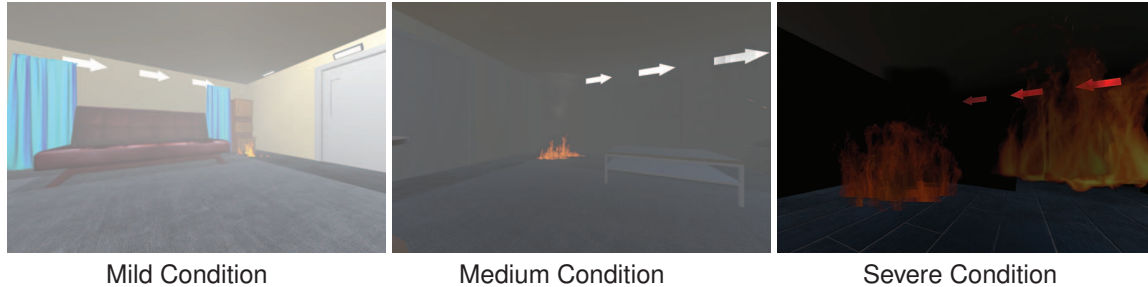


Figure 1: Example screenshots from three dangerous conditions in our virtual evacuation study where participants need to choose the left or right door to exit. The AI augmentation is provided through both visual (flashing arrows) and audio instructions.

## ABSTRACT

While Artificial Intelligence (AI) has quickly proliferated, the understanding of trust towards AI is still under-researched and it is a primary factor for the success of AI-augmentation. This project created a virtual evacuation study to explore people's trust of AI in dangerous situations. Our study adopts a virtual maze-like scenario with automatic AI-augmentation and three levels of fire simulations, where participants need to make the final decision. We have collected performance data and revised the metrics of KUSIV3 for participants to report their trust of AI in the post-questionnaires. Our results show that the levels of trust towards AI are generally positive, and can be affected by dangerous conditions. The participant-reported and measured trust levels also demonstrate a loose but consistent correlation. We plan to expand this study to design effective AI-augmentation interfaces to support decision-making procedures.

**Index Terms:** Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality; Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

## 1 INTRODUCTION

AI augmentation tools are becoming powerful and widely used. The future of AI can help humans collect information automatically and provide intelligent suggestions at various work and life situations [4]. One especially useful application is evacuation, where people can often only observe local conditions, while AI can combine information from a wide area to provide optimized suggestions.

However, human's trust on AI may differ, which can affect the final results significantly [8]. It is important to understand the trust effects, so that we can design trustful AI systems in order to adapt to individual variances on decision making procedures under different conditions. Previous work on studying trust in virtual environments include measuring trust in virtual agents [3, 5], in Human-Machine teams [6], and in AI assistant [2]. This problem is still under-researched and it is very challenging to capture such internal factors that cannot be modeled simply [4].

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This work studies the trust problem in an emergent evacuation situation. Our design is similar to the virtual maze [5], where users are placed in a room with two doors and they need to decide on which door to exit. Conversely, we are interested in exploring the trust of humans towards AI when people are in danger, so we show the AI augmentation with both visual and sound cues and create dangerous fire situations at three levels. As shown in Figure 1, the visual cues are the three large flashing arrows that indicate the suggested direction, and the auditory cues consist of "evacuate to the left" or "evacuate to the right". We specifically inform all participants that 1) the suggestions provided in the study are from AI and 2) they need to make the final decision.

We analyzed the trust effects with data collected from the user study and revised the Interpersonal Trust Short Scale (KUSIV3) [1] for AI in the post-questionnaires. While it is a basic measurement, our result show a consistent correlation between self-reported and measured trust levels. Our results also demonstrate that the trust levels from majority participants are positive across all dangerous conditions, and AI augmentation may be the most helpful when people are alert but not in panic under the medium condition.

## 2 METHOD

### 2.1 Virtual Evacuation Environment

The environment created is a standard rectangular room with two doors at the center on the left and right side of the room. This room size and door locations are kept the same for all the trials, and we adjust the inside with variable furniture and lighting. To create emergent situations, we have developed fire and smoke simulations and randomly added to all the trials. Since we focus on the condition of danger, our study contains three dangerous levels – mild with two or three places on small fires and clear visibility, severe with several places on fires and heavy dark smokes, and medium in between. We have balanced the locations of fires and smokes on both sides of the room to create fair comparisons among all the trials. To create realistic simulations, we also added sounds of fires, coming ambulance, and people chaos noises on the background. The sound effects are also increased with the dangerous level.

### 2.2 Participants

We have recruited 10 participants, 5 females and 5 males, all between the ages of 20 and 50. All the participants major in Computer Science and have previous VR experiences. All participants had a normal or corrected-to-normal vision. The study was approved by the IRB committee at our institution.

### 2.3 Procedure

The participants are first given an introduction to the study. We inform all participants that though an AI can give them suggestions, it cannot make decisions for them. Before the trials begin, they are given a practice session that has no bearing on the data. Then, our VR program takes the participants through the 12 trials with 4 trials for each of the 3 dangerous conditions. We mixed the order of danger conditions to avoid any sequential effect on the study. There is a rest stage after each trial, and participants can choose to continue the study whenever they are ready.

After completing all the trials, we asked participants to remove the HMD, take post-questionnaires and comment on their decisions. We used the Simulator Sickness Questionnaire (SSQ) [7] to measure the general VR experience. Since we are interested in the trust to AI, we revised the Interpersonal Trust Short Scale (KUSIV3) [1] for AI as follows: “In the following we are interested in your attitude towards AI. Please indicate the extent to which you agree or disagree with each of the statements below.”

(Q1) I am convinced that most AI algorithms have good intentions.

(Q2) You cannot rely on any AI algorithms these days.

(Q3) In general, AI can be trusted.

The response specifications use the same 5-point rating scale.

### 2.4 Measures

We collected the performance records from all the participants, including the decisions participants made and the duration spent for each trial. We also summarized the responses to the post-questionnaires. For the questions of trust on AI, we use the equation  $Q1 + Q3 - Q2$  as the self-reported trust level.

### 3 DATA ANALYSIS AND RESULTS

To explore the factor of dangerous condition, we first measure the completion time of all the trials. As shown in Figure 2 (left), the average completion time increases with the degree of danger, although this is not statistically significantly. The standard deviations of the time from individual participants ranges from 1.23 to 4.89. The large values suggest different decision-making strategies that are also been observed in the comments, such as “choose the direction with lesser fires”, “escape by staying away from the closest fires”, and “too dark to observe clearly, so just follow the suggestion”.

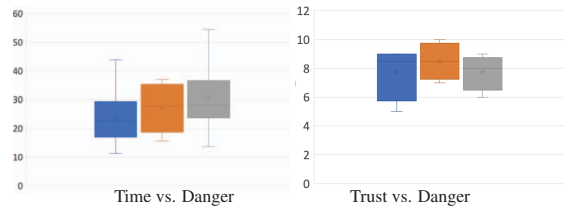


Figure 2: The columns represent mild, medium and severe dangerous conditions. The completion time increases with the dangerous conditions. The measured trust levels are similar and the highest average value is on the medium condition.

To measure trust levels, we first explore a basic method by comparing the decision of participants with the AI suggestions. As the Figure 2 (right) shows, the average trust levels for the three danger conditions are 65%, 71% and 65%. Four out of the 12 participants obeyed the AI suggestions more than 90% of the time. The medium danger condition corresponds to the highest average value. Medium danger is likely when AI is most useful, in a place between a good human decision under mild conditions and a snap decision based on panic under a severe condition.

We also summarized the answers from the trust questionnaires. Figure 3 also shows a loose correlation ( $R = 0.663$  and  $STD = 2.00$ ) between the measured trust levels and the self-reported answers.

This is after removing one data point (2,12), as it is an outlier according to the Cook’s Distance ( $0.7 > \text{threshold } 4/n = 0.4$ ).

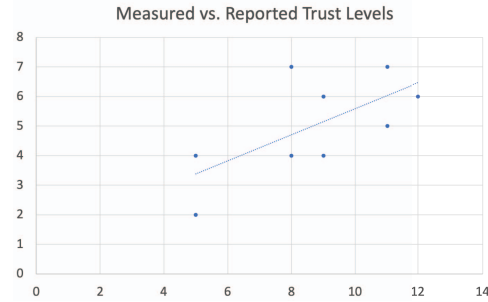


Figure 3: The reported levels are from the responses to post-questionnaires, and the measured levels are from the study results.

All the participants acknowledged that the visual and sound cues can be easily recognized. The results from SSQ questionnaires show no major discomfort from all the participants.

### 4 CONCLUSION AND FUTURE WORK

This project designs a virtual evacuation study to explore a fundamental problem of AI-augmentation – trust of AI. Our positive results on the trust levels from majority participants and trials support the methods of AI-augmentation for saving people under dangerous conditions. There is also a great potential to measure the individual differences on the trust of AI through both questionnaires and data analysis of user behaviors. Our study can be expanded to explore additional user behaviors with AI augmentation in different situations, and help researchers to design AI-augmenting interfaces to effectively support decision-making procedures.

### ACKNOWLEDGMENTS

This work was supported in part by NSF grant 1840080.

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